Dilpreet Singh Bansi

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EDUCATION

York University **Toronto, ON**

B.Eng. in Software Engineering · GPA 3.7/4.0 · Dean's List (2021–2026), Honour Roll, Entrance Scholarship *May 2026*

- **Coursework:** Machine Learning, Distributed Systems, Data Structures, Computer Networks, Software Development

EXPERIENCE

SOTI **Mississauga, ON**

Software Developer, AI/ML *May 2026 – Present*

- Led natural-language app generation in SOTI Snap, the company's enterprise low-code platform, turning plain-English descriptions of screens, forms, and workflows into production-ready iOS and Android apps and cutting app build time 80% for business users
- Architect LLM pipelines combining RAG, prompt engineering, and agentic workflows that turn requirements into working UI, data models, and backend logic across 100+ languages
- Partnered 1:1 with the CEO and executives on the AI product roadmap; shipped a working prototype in the first month to validate the lead feature with real users
- Ship AI features in C#/.NET and Angular under Agile/Scrum, backed by automated unit and integration tests, performance benchmarks of response latency and token cost, and LLM output evaluation against reference prompts and expected outputs

Tegra Lumber **Cambridge, ON**

Software Engineer, Full-Stack *Jan 2024 – May 2026*

- Developed the company's operations platform end to end as sole engineer, starting with inventory and yard management modules on React/TypeScript, Node/Express, and PostgreSQL that tracked stock by yard location; cut cycle-count time 45% and stockouts 28%
- Engineered a pricing and quoting service combining board-foot conversion, tiered and contract pricing, and fuel surcharges for every contractor order, cutting quote turnaround from 6 hours to 5 minutes
- Launched a customer order portal showing real-time availability, pricing, and pickup scheduling; moved 14% of counter orders online within 6 weeks and cut peak-hour phone traffic 35%
- Raised test coverage to 82% with Jest unit tests and Playwright end-to-end tests, catching 14 regressions before release across 14 releases; reached 93% adoption among staff new to software
- Set up CI/CD (GitHub Actions, Docker, automated migrations) with blue-green deploys; cut deploy time from 15 to 2 minutes

PROJECTS

Llama-wgsl

JavaScript, WebGPU, WGSL

- Ran SmolLM2 on the user's GPU inside the browser with hand-written WGSL compute kernels for fused QKV and SwiGLU matvecs, grouped-query attention, RoPE, and the KV cache; it needs no server or dependencies, so the page deploys as static files
- Streamed safetensors weights to GPU memory as packed bf16 and reused cached prefixes across chat turns; matched Hugging Face transformers logits within 5e-5 and greedy output token for token

Yardline

TypeScript, React, Node.js, PostgreSQL

- Developed a lumber-yard quoting and ordering app that parses messy material lists (34 formats), fuzzy-matches a catalog, prices by board foot with tier discounts, and pushes live updates to every open tab over SSE
- Prevented lost updates with version checks and overselling with row locks, made orders idempotent, and wrote 115 tests including racing Postgres transactions; the live demo runs Postgres (WebAssembly) in the browser

FlashServe

Python, PyTorch, FastAPI

- Built a Llama inference engine for Hugging Face checkpoints with FlashAttention-style tiled attention, KV-cache decoding running 6.1× faster than recompute, speculative decoding, and a paged KV cache, exposed through an OpenAI-compatible streaming API

TinyTrain

Python, PyTorch

- Implemented data, tensor (Megatron) and pipeline (GPipe) parallelism from scratch, splitting data batches, weight matrices and model layers across separate GPU workers, and tested each to match single-process training on losses and gradients at every step

NeuronLens

Python, PyTorch

- Built an interpretability toolkit that independently reproduced published GPT-2 circuits: induction heads, indirect object identification name movers, and causal tracing of factual recall

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, C#, C++, SQL, Java, WGSL
AI/ML: PyTorch, Hugging Face, LLM inference, RAG, agents, evals, LangChain/LangGraph, OpenAI API, WebGPU
Backend & Cloud: FastAPI, Node.js, .NET, PostgreSQL, Redis, MongoDB, Docker, Kubernetes, AWS, GCP, CI/CD
Frontend & Testing: React, Angular, HTML/CSS, Jest, Vitest, Playwright, Pytest